"What Were They Thinking?" Using Open-Text Responses to Validate Constructs in Survey Experiments

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Abstract

Survey experiments enable causal inference, but researchers rarely verify whether respondents process experimental stimuli as intended. This paper introduces a two-pronged framework for construct validation using open-text responses embedded in survey experiments, demonstrating two complementary approaches: structural topic modeling (STM) to understand responses using unsupervised topic modeling, and large language model (LLM) classification to confirm whether responses align with theoretically expected constructs. Applying this framework to conjoint experiments on immigrant naturalization in South Korea ($N = 1{,}999$) and Taiwan ($N = 2{,}050$), I find that a social integration cue shifts respondent reasoning, in both countries, from economic and practical criteria toward civic engagement themes. Two independent methods converge: data-driven STM identifies civic topics that increase 7–11 percentage points when the cue is shown, while theory-driven LLM classification shows civic commitment rising 15–18 percentage points. Six independent LLM classifiers (Fleiss' $\kappa = 0.745$), spanning 3B to 72B parameters, produce consistent results. A second application to a standard conjoint on immigrant admission demonstrates that the framework transfers to the more common experimental setup, and validation against expert human-coded ground truth from a third conjoint confirms that iterative codebook refinement produces substantial LLM-human agreement.

Keywords: survey experiments, construct validation, open-ended questions, structural topic models, large language models, text classification, immigration attitudes

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Introduction

Survey experiments are among the most common designs in contemporary social science. Random assignment to treatment conditions allows researchers to estimate causal effects on attitudes, preferences, and behaviors. Yet an assumption underlying these designs often goes untested: that respondents process experimental stimuli as intended and that closed-ended response items capture the constructs researchers claim to measure.

This assumption is especially tenuous in conjoint experiments [Hainmueller et al., 2014], where respondents evaluate multi-attribute profiles. A conjoint profile may contain ten or more randomly varied attributes, but researchers typically interpret treatment effects as if respondents attended to and processed each attribute independently. The possibility that respondents focused on entirely different aspects of the profile, or interpreted attributes in unexpected ways, is rarely examined.

I propose a straightforward solution: embed open-text questions in survey experiments and use computational text analysis to systematically examine respondent reasoning. When a respondent is asked "Why did you choose this person?" after a conjoint task, the answer shows the reasoning behind the observed choice. If the experimental manipulation shaped respondent reasoning as intended, this should be visible in the open-text responses.

I develop and demonstrate a two-pronged analytical framework. The first approach is structural topic modeling (STM) for *discovery*: what topics did respondents discuss, and how did experimental conditions affect topic prevalence? STM is well-suited to this task because it incorporates treatment conditions as covariates during model estimation [Roberts et al., 2014]. The second is large language model (LLM) classification for *confirmation*: do open-text responses align with theory-derived categories? Following recent advances in codebook design for LLM measurement [Halterman and Keith, 2025], I classify responses into substantive codes that correspond to the constructs the experimental design aims to activate.

The empirical application uses original conjoint survey data on attitudes toward immigrant naturalization in South Korea ($N = 1{,}999$) and Taiwan ($N = 2{,}050$), collected in 2024, extending conjoint research on immigration attitudes [Hainmueller and Hopkins, 2015, Denney and Green, 2021]. The conjoint design includes a "social integration" attribute that randomly describes immigrants' civic engage-

ment activities. The validation question is direct: when respondents see the social integration cue, do they actually write about civic engagement?

They do. In both countries, the social integration cue increases discussion of civic and community themes while reducing discussion of economic and credential-based criteria. These effects range from 7 to 18 percentage points depending on the method and country, and they hold across both analytical approaches and six diverse LLM classifiers (Fleiss' $\kappa = 0.745$), providing evidence of substantive validity for the "cues of commitment" framework underlying the conjoint design. To be precise, this is one form of validity evidence (that experimental stimuli shift the content of respondent reasoning in the predicted direction) rather than full construct validation in the psychometric sense, which would require convergent and discriminant evidence across multiple operationalizations.

The contribution is primarily methodological. I develop a two-stage workflow for construct validation: unsupervised topic modeling first discovers the empirical structure of open-text responses, and those findings then inform the design of a theory-driven codebook for LLM classification. Unlike approaches that apply LLM classification or scaling directly to text [Mellon et al., 2024, DiGiuseppe and Flynn, 2026], this workflow uses unsupervised findings to check and refine the codebook before classification begins. In this case, the initial codebook conflated two constructs that the STM showed respond differently to the experimental treatment; the revised version separated them, and the resulting classification was sharper for it. Validation against expert human-coded ground truth from a separate conjoint experiment confirms that iterative codebook refinement brings LLM-human agreement to $\kappa \approx 0.6$–0.7. All classification is done in six models, one proprietary (GPT-4o-mini) and five open-weight from 3B to 72B parameters, to test reproducibility across model families and sizes [Barrie et al., 2025, Chae and Davidson, 2025].

Background and Related Work

Open-Ended Questions in Survey Research

Open-ended questions have a long history in survey research, but their systematic use for construct validation is less established. Geer [1988] showed that open-ended responses capture genuine attitudes rather than mere articulateness. Lupia [2018] documented persistent quality problems in coding open-ended responses and called for greater transparency. Glazier et al. [2021] introduced a "self-coding" method where respondents categorize their own open-ended responses, a direct check on whether researcher-imposed categories match respondent interpretations. Ziegler [2022] uses text-as-data methods on open-ended survey responses as manipulation checks. Haaland et al. [2025] review the growing use of open-ended survey data to uncover mechanisms behind beliefs and behavior, including LLM-based annotation at scale.

This paper extends that line of work by using computational text analysis to scale from small hand-coded samples to entire survey populations.

Structural Topic Models

Topic models provide a data-driven approach to discovering the thematic structure of text corpora [see Grimmer and Stewart, 2013, Grimmer et al., 2022, Gentzkow et al., 2019].[1] Roberts et al. [2014] developed the Structural Topic Model for contexts where document-level metadata (such as experimental treatment conditions) should inform topic estimation; Roberts et al. [2016] provide the formal statistical foundation. Unlike standard LDA, STM allows covariates to affect both topic prevalence (which topics appear in which documents) and topic content (which words are associated with which topics). This makes STM ideal for survey experiments: researchers can directly estimate how experimental conditions affect the topics respondents discuss [see also Egami et al., 2022, Fong and Grimmer, 2023, on causal inference with text].

[1] For the foundational Latent Dirichlet Allocation model, see Blei et al. [2003].

Hobbs and Green [2025] distinguish between topic discovery and attitude inference, arguing that topic models excel at the former but require careful interpretation for the latter. The present application is primarily concerned with topic discovery (identifying what respondents discussed) and with treatment effects on topic prevalence.

LLM-Based Text Classification

Large language models are now widely used for text classification [Gilardi et al., 2023, Ziems et al., 2024, Bail, 2024]. Chae and Davidson [2025] compare zero-shot, few-shot, and instruction-tuned LLMs for text classification in social science, finding that model size and prompting strategy jointly determine classification quality. The present study tests this directly by deploying models from 3B to 72B parameters. Törnberg [2025] demonstrates that LLMs can outperform human coders on contextual reasoning tasks. Heseltine and Clemm von Hohenberg [2024] show that hybrid human-LLM workflows achieve 93%+ accuracy cost-effectively. Ornstein et al. [2025] provide the `prompt` R package for few-shot LLM classification. Mellon et al. [2024] demonstrate that LLMs can code open-text survey responses at scale with accuracy comparable to human coders, while Rathje et al. [2024] show that GPT performs well for text analysis across 12 languages, supporting multilingual classification. Most of this work assigns each response to a discrete code. DiGiuseppe and Flynn [2026] take a different tack, using LLM pairwise comparisons and Bradley-Terry modeling to place responses on a continuous scale. The choice between categorical and continuous measurement depends on the research question; the present study asks what *kind* of reasoning respondents use, so categorical codes are the natural fit.

Codebook design matters more than model choice. Halterman and Keith [2025] argue that the quality of LLM classification depends on the precision of the codebook: loose instructions lead to poor performance regardless of model capability. They recommend codebooks with explicit definitions, inclusion criteria, exclusion criteria, and worked examples.

Reproducibility is an equally pressing concern. Barrie et al. [2025] show that proprietary LLM APIs exhibit substantial variance across runs and model versions, threatening the reproducibility of research findings. They recommend open-weight models that can be run locally with fixed parameters, producing

deterministic outputs [see also Le Mens and Gallego, 2025]. This concern motivates the present study's design: five open-weight models form the reproducible core, while GPT-4o-mini is included as a reference point for the proprietary models that dominate current practice in political science. Gilardi et al. [2023] used GPT-3.5-turbo, Törnberg [2025] and Heseltine and Clemm von Hohenberg [2024] used GPT-4, and Chae and Davidson [2025] used GPT-4o. The open-weight results show that researchers need not rely on proprietary APIs to achieve comparable classification quality.

Data and Research Design

The Conjoint Experiment

The data come from original surveys conducted in South Korea and Taiwan between January and June 2024, examining public attitudes toward immigrant naturalization [Denney et al., 2025]. Respondents were recruited via Qualtrics using stratified quota sampling by sex, age, and geographic region to approximate national census distributions. The final samples comprise $N = 1{,}999$ in South Korea (51% male) and $N = 2{,}050$ in Taiwan (53% male).

Each respondent evaluated paired immigrant profiles in a forced-choice conjoint experiment [Hainmueller et al., 2014, Bansak et al., 2021]. In South Korea, respondents completed 10 paired-choice tasks; in Taiwan, 7 tasks. Profiles varied on 12 randomly assigned attributes covering demographic characteristics, economic qualifications, and social integration (see Supplementary Information, Table A.2 for the full attribute table).

The attribute of interest for this analysis is *social integration*, which describes the immigrant's civic engagement activities with four levels: "only associates with fellow expatriates," "often spends time alone," "co-hosts monthly multicultural events," and "volunteers weekly at a local community center." Due to the randomization design, this attribute was randomly withheld from some profiles, creating a clean cue-shown vs. cue-not-shown comparison.

After the conjoint task, respondents answered an open-text question: *"Why did you choose this person as more deserving of citizenship?"* The question appeared once, after the final conjoint choice, and was

required (respondents could not skip it). Responses ranged from single words to several sentences. No minimum length was imposed, and no responses were excluded on the basis of brevity alone.

Open-Text Corpus

The corpus comprises 4,049 responses: 1,999 from South Korea (Korean) and 2,050 from Taiwan (Traditional Chinese). Responses are brief (median: 10 characters in Korean, 6 in Chinese) but substantive. Based on majority-vote LLM classification, approximately 79% of responses receive a substantive code (civic commitment, functional integration, identity concern, or economic focus), indicating that the majority contain identifiable references to conjoint attributes.

The treatment variable is `cue_shown`: whether the social integration attribute was displayed on the conjoint profile the respondent evaluated. Due to the attribute randomization design, approximately half of responses in each country come from profiles where the social integration cue was shown (South Korea: 49.8%; Taiwan: 49.1%).

Analytical Framework

Approach 1: Structural Topic Modeling

I fit separate STM models for each country, using the `stm` package in R [Roberts et al., 2019] with spectral initialization and seed = 42. Korean text is preprocessed using the `kiwipiepy` morphological analyzer and Chinese text using `quanteda`'s built-in segmenter [Benoit et al., 2018]. Preprocessing follows established guidelines for multilingual text analysis [Lucas et al., 2015, Denny and Spirling, 2018].

Since the conjoint design is fully randomized, the prevalence specification includes only the treatment indicator:

$$\text{prevalence} \sim \texttt{cue_shown} \tag{1}$$

I select the number of topics (K) using the coherence-exclusivity tradeoff from `searchK()` diagnostics, resulting in $K = 6$ for South Korea and $K = 8$ for Taiwan.

Treatment effects on topic prevalence are estimated using `estimateEffect()`, which provides coefficient estimates, standard errors, and confidence intervals for the effect of `cue_shown` on each topic's prevalence.

Approach 2: LLM Classification

I developed the codebook iteratively, using STM findings to guide refinement. The initial version (v1) used four codes: `civic_commitment`, `identity_concern`, `economic_focus`, and `other`. The civic commitment code captured civic engagement, volunteering, and community involvement, but also included language competence and length of residence as signals of integration commitment.

Running v1 across GPT-4o-mini exposed a problem: 561 responses (13.9%) were coded as civic commitment based solely on language fluency or residence duration, which are practical attributes, not active civic engagement. The STM results showed that topics about language and residence *decrease* when the social integration cue is shown, meaning these attributes behave as substitutes for civic engagement, not signals of it. Including them under civic commitment conflated participation in community life with individual-level adaptation to the host country.

This STM-informed diagnosis led to a revised codebook (v2) with five codes:

Code	Definition
`civic_commitment`	Participation in community life: volunteering, community organizing, social participation. Outward-facing, prosocial engagement
`functional_integration`	Individual-level adaptation: language competence, residence duration, law-abiding behavior. Practical prerequisites for functioning in the host country
`identity_concern`	Ethnic, cultural, or national identity concerns, including religion and ethnicity when used as evaluation criteria
`economic_focus`	Economic contributions: jobs, taxes, career, occupation, work experience
`other`	Truly uninformative responses: filler words, gibberish, meta-comments. Brevity alone is never a reason for this code

Table 1: LLM classification codebook (v2). Codes follow the design principles of Halterman and Keith [2025], with explicit definitions, inclusion/exclusion criteria, and few-shot examples in both Korean and Traditional Chinese.

The classification pipeline uses six LLMs. GPT-4o-mini (OpenAI API) provides a reference point for the proprietary models used in prior work. Five open-weight models form the reproducible core:

Model	Parameters	Infrastructure	Role
GPT-4o-mini	—	OpenAI API	Proprietary baseline; representative of models used in current practice
Qwen 2.5 3B	3B	Consumer GPU	Open-weight; smallest model
Llama 3.1 8B	8B	Consumer GPU	Open-weight; mid-range
Gemma 3 12B	12B	University HPC (A100)	Open-weight; mid-range
Qwen 2.5 32B	32B	University HPC (A100)	Open-weight; large
Qwen 2.5 72B	72B	University HPC (A100)	Open-weight; largest model; size scaling

Table 2: LLM classifiers used in the analysis. Open-weight models are run via the Ollama framework with temperature = 0 and seed = 42 for deterministic output.

All open-weight models are run with temperature = 0, seed = 42, and max_tokens = 200, producing deterministic outputs. GPT-4o-mini is run with the same parameters, but proprietary API outputs are not strictly reproducible across runs due to batched inference and silent model updates [Barrie et al.,

2025]. The open-weight models therefore provide the reproducibility guarantee; GPT-40-mini serves as a reference point for the proprietary models that dominate current practice. The range of model sizes (3B to 72B parameters, including three from the Qwen 2.5 family) allows a direct test of whether classification quality scales with model size [Chae and Davidson, 2025].

The classification pipeline is implemented in R and Python. Codebook generation (from a structured YAML definition following Halterman and Keith 2025), cross-model validation, and all statistical analyses are written in R; the OpenAI API classifier is a standalone Python script. Open-weight models are deployed through Ollama, a lightweight framework for running LLMs locally with fixed inference parameters. The two smallest models (3B, 8B) run on a consumer GPU (NVIDIA RTX 3060 Ti, 8GB VRAM); the three larger models (12B, 32B, 72B) run on a university high-performance computing cluster equipped with NVIDIA A100 GPUs, with the 32B and 72B models quantized to 4-bit precision to fit within GPU memory. All code, codebooks, and classification outputs are available in the replication archive.

Each response is classified independently by all six models. Because no human-coded ground truth exists for these responses, the analysis relies on cross-model consensus rather than any single model's judgment. The majority-vote classification assigns each response the code that four or more of the six models agree on; responses without a four-model consensus (7.2% of the corpus) are excluded. This approach treats the six LLMs as diverse raters (varying in architecture, size, and training data, though not fully independent given shared pretraining corpora) and applies inter-rater reliability standards conventional in content analysis [Benoit et al., 2026]. Because the models share pretraining data, agreement statistics likely overstate what fully independent annotators would produce. The ground-truth extension, where LLM-human agreement reaches $\kappa \approx 0.6$–0.7, provides a more realistic lower bound on classification quality than the inter-model κ of 0.745. Majority-vote aggregation reduces the influence of model-specific biases (for example, Llama 3.1 8B's inflated functional integration rate in Taiwan) by requiring broad agreement across model families and sizes before any classification is assigned. Treatment effects are then estimated as differences in code prevalence between cue-shown and cue-not-shown conditions on this consensus-classified corpus, with bootstrap confidence intervals (2,000 iterations, percentile

method) providing inference.

The individual models produce consistent results even without aggregation. Table C.6 in the Supplementary Information reports the cue effect for each model separately: all six agree on the direction of every substantive effect, with civic commitment increasing between +9.3 and +18.1 percentage points across all models and countries. Even the smallest model (Qwen 2.5 3B) produces treatment effect estimates in the correct direction for every code in every country. The majority-vote procedure is thus a conservative safeguard, not a correction for disagreement. Results are stable across alternative thresholds (3/6, 5/6) and when GPT-4o-mini is excluded entirely (Supplementary Information, Appendix E). A simple keyword/dictionary baseline detects the same treatment direction but attenuates the magnitude by half ($\kappa = 0.51$ with the LLM majority vote), which shows that LLMs outperform keyword matching on this task.

Triangulation

The two approaches answer different questions. STM shows the empirical structure of the text without imposing prior categories. It can surface patterns the researcher did not anticipate. But STM cannot tell you whether a topic corresponds to a specific theoretical construct, and on very short text (median 6–10 characters), topic assignments are often diffuse. LLM classification maps responses onto theory-derived categories with greater precision, but only if the categories are right. An ill-specified codebook produces precise but misleading estimates.

Running both methods on the same data is useful precisely when they disagree. The STM results in this study showed that language and residence topics move in the *opposite* direction from civic engagement topics under the treatment. That empirical pattern, which emerged without any prior categorization, made clear that the v1 codebook was grouping together constructs that respond differently to the experimental manipulation. LLM classification alone would not have flagged this problem, because the model faithfully follows whatever codebook it is given. The STM provided the evidence needed to refine the measurement before drawing conclusions from it.

This is not triangulation in the sense of independent data sources confirming the same finding; both

methods operate on the same text corpus. It is a workflow in which each method has a specific role: STM identifies the empirical structure, LLM classification applies theoretical categories informed by that structure, and the combination catches errors that either method alone would miss. The key difference from single-stage LLM classification is the feedback loop: the unsupervised step diagnoses problems in the codebook that the supervised step, on its own, would faithfully reproduce.

A structural difference between the two methods also explains why they produce different effect magnitudes. STM assigns every document a probability distribution over all topics, including uninformative responses. Noise gets distributed across the topic space rather than isolated, producing conservative treatment effect estimates that reflect the full corpus [Roberts et al., 2014]. LLM classification includes an explicit `other` category that filters uninformative responses out of the substantive codes, so treatment effects are measured against a cleaner denominator. The full-sample STM estimates are not wrong; they reflect a different estimand. The Supplementary Information (Appendix B) provides a sensitivity analysis showing that as diffusely assigned STM documents are progressively excluded, the STM civic effect increases monotonically and converges toward the LLM estimate.

Results

STM: What Did Respondents Write About?

The South Korean model ($K = 6$) identifies six topics that map onto conjoint profile attributes and respondent evaluation criteria. The largest topics concern identity and practical credentials: religious/ethnic preferences (22.9%), residence and assimilation effort (21.3%), and community engagement and volunteering (18.8%). Employment and family stability (13.1%), professional experience (12.1%), and language/visa considerations (11.9%) account for the rest. Taiwan's model ($K = 8$) produces a comparable structure, with language fluency and first impressions (18.6%), work experience and social integration (18.5%), and civic activities and ethnicity (14.6%) as the largest topics.

That respondents wrote about the attributes they saw in the immigrant profiles is itself a finding. It means they processed the experimental stimuli and evaluated candidates on the dimensions the conjoint

design intended to test.

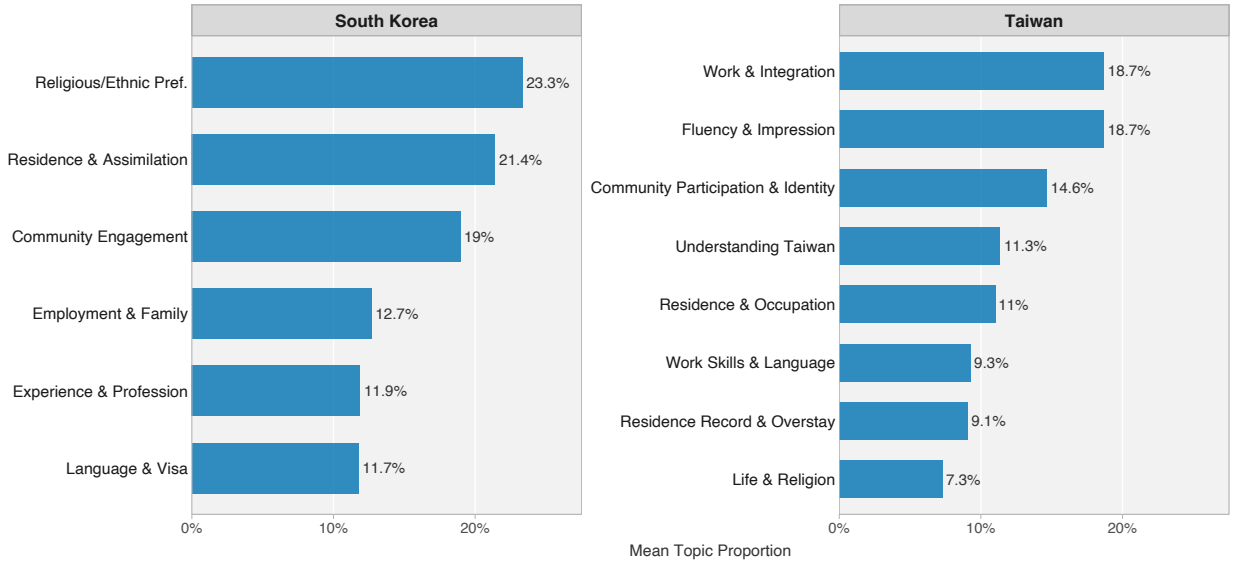

Figure 1: STM topic prevalence by country. Mean document-topic proportions for South Korea ($K = 6$) and Taiwan ($K = 8$).

STM: Effect of the Social Integration Cue

In South Korea, the social integration cue increases discussion of Community Engagement by 6.7 percentage points while decreasing discussion of Language & Visa (-4.0 pp) and Employment & Family (-2.3 pp). All three effects are statistically significant.

Taiwan shows the same pattern at greater magnitude. The cue increases Community Participation & Identity by 11.4 pp, Work & Integration by 3.1 pp, and Understanding Taiwan by 1.8 pp, while decreasing Fluency & Impression (-5.4 pp), Residence & Occupation (-4.7 pp), Residence Record (-3.0 pp), and Work Skills (-2.7 pp). All effects are significant at the 95% level; see Figure 2.

The pattern is consistent across both countries, and the *substitution* effects are as informative as the civic increase. If the open-text responses merely reflected attribute recall, respondents parroting back what they saw, we would expect the civic cue to add civic language without displacing other themes. Instead, language, residence, and employment topics all decline, suggesting that the cue reshapes the evaluative framework rather than adding a talking point. This substitution pattern is the strongest evidence that the social integration attribute affects how respondents reason about the profiles, not just what they

mention afterward.

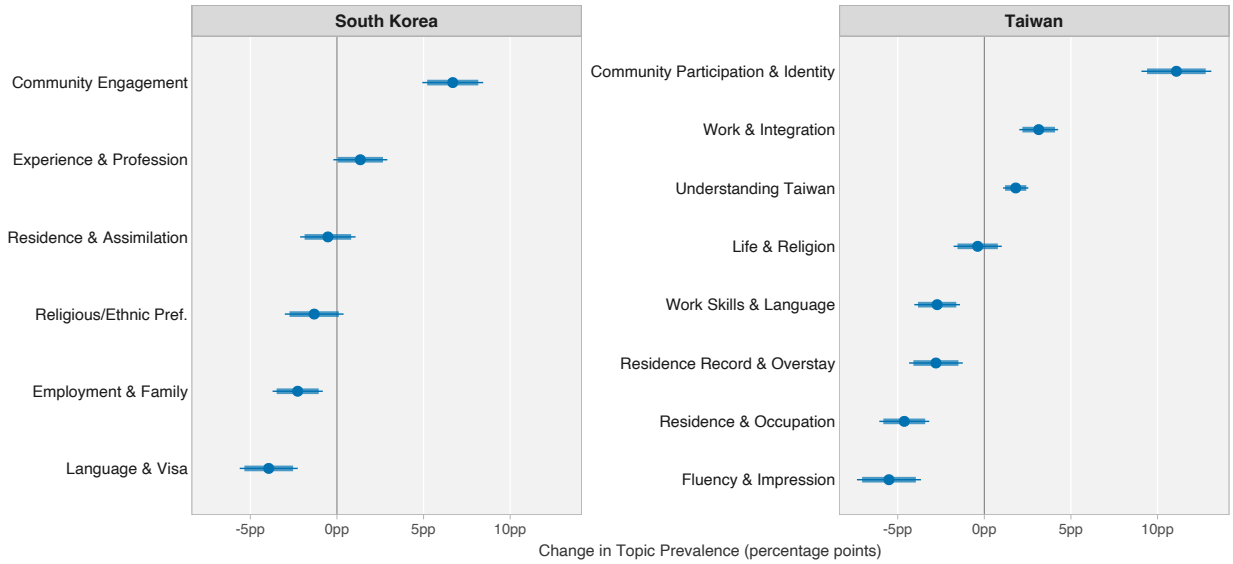

Figure 2: Effect of social integration cue on topic prevalence. STM coefficient estimates with 90% (thick) and 95% (thin) confidence intervals.

LLM Classification: Code Prevalence

Using majority-vote classification (4+ of 6 models agree), functional integration is the most common substantive code in both countries (∼32%), followed by identity concern (15–21%), economic focus (19–23%), and other (20–21%). Civic commitment (active behavioral engagement) accounts for 2.0% (South Korea) and 1.0% (Taiwan) when the cue is *not* shown, rising to 16.6% and 19.2% when it *is* shown.

Separating functional integration from civic commitment makes clear that practical attributes (language, residence, law-abiding behavior) dominate respondent reasoning in both countries, distinct from the active civic engagement that the cue is designed to activate.

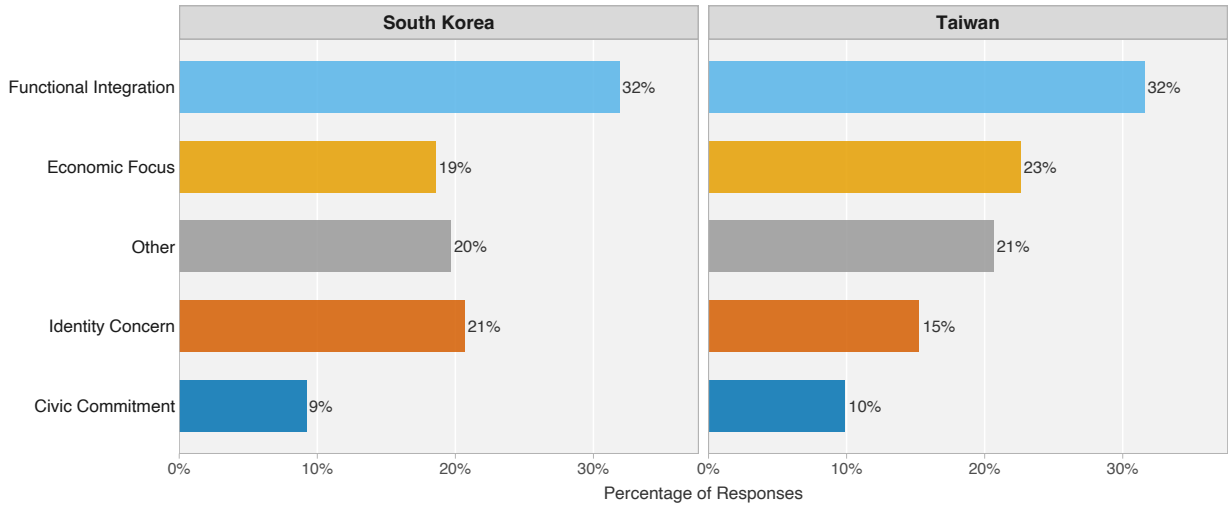

Figure 3: LLM classification: code prevalence by country. Majority-vote classification (4+ of 6 models agree).

LLM Classification: Effect of the Social Integration Cue

In South Korea, the cue increases civic commitment by +14.7 percentage points (from 2.0% to 16.6%), while decreasing functional integration by -7.9 pp and economic focus by -5.6 pp. Identity concern decreases modestly (-2.7 pp).

The Taiwanese results are similar in direction but stronger in magnitude: civic commitment increases by +18.2 pp (from 1.0% to 19.2%), while economic focus decreases by -11.4 pp and functional integration by -6.8 pp. Identity concern is stable (-1.0 pp).

All prevalence figures and treatment effects include the `other` category in the denominator. Table C.7 in the Supplementary Information reports the same effects computed among substantive codes only (excluding `other`), which produces larger point estimates.

Bootstrap confidence intervals place all of these effects outside zero at the 95% level. Across the six individual models, civic commitment effects range from +9.3 to +16.1 pp in South Korea and +14.4 to +18.1 pp in Taiwan, all significant. The treatment effect is present across all response lengths, including responses of five characters or fewer ($+7$ pp), though it is larger for longer responses (Supplementary Information, Appendix E).

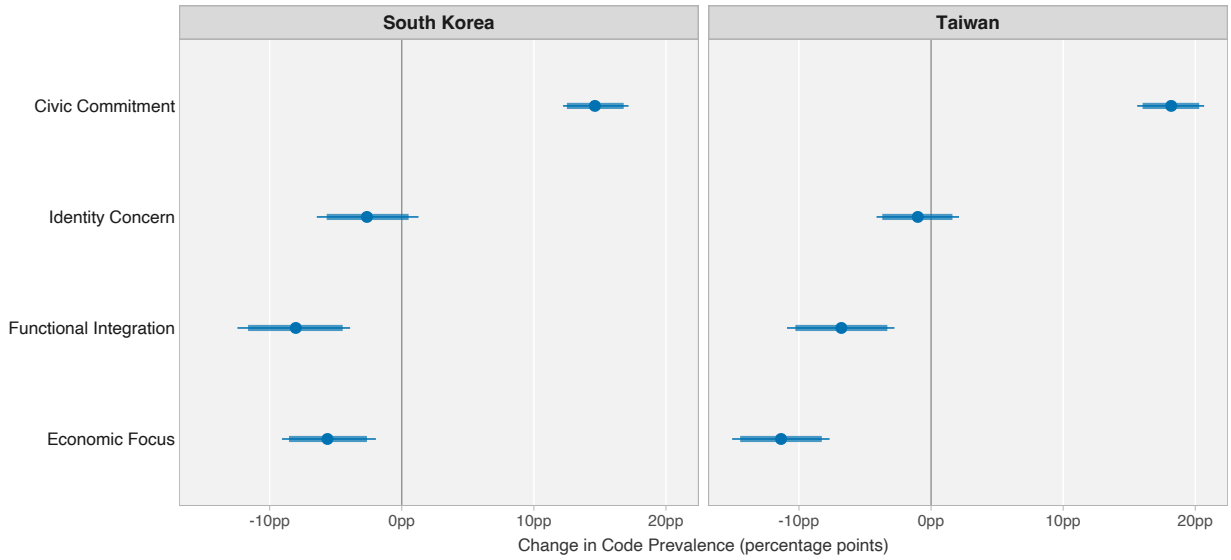

Figure 4: Effect of social integration cue on LLM-classified response themes. Majority-vote classification (4+ of 6 models). Bootstrap 90%/95% CIs (2,000 iterations).

All six LLMs produce consistent cue effect estimates (Figure 5; see also Table C.6 in the Supplementary Information for the full per-model breakdown). All six models agree that the social integration cue significantly increases civic commitment and decreases economic focus in both countries. Pairwise agreement ranges from 66.3% to 91.2% (Cohen's κ: 0.578–0.886), with Fleiss' multi-rater $\kappa = 0.745$ indicating substantial agreement.

Agreement increases with model size. Within the Qwen 2.5 family, pairwise κ with GPT-4o-mini rises monotonically: 0.744 (3B), 0.841 (32B), 0.860 (72B). The highest agreement overall is between Qwen 2.5 32B and 72B ($\kappa = 0.886$, 91.2% raw agreement), and the strongest cross-family pair is GPT-4o-mini and Qwen 2.5 72B ($\kappa = 0.860$, 89.1%). Larger open-weight models converge toward each other and toward the proprietary model. Chae and Davidson [2025] report a similar pattern.

Overall, 92.8% of responses receive a majority classification (4+ of 6 models agree). The weakest performer is Llama 3.1 8B, which inflates functional integration classifications in Taiwan (43.7% vs. 30–37% for the four larger models), probably due to weaker CJK language capabilities. Yet even this model produces civic commitment effects in the correct direction (+12.2 pp in South Korea, +15.9 pp in Taiwan), and the smallest model in the study (Qwen 2.5 3B) agrees on the direction of every effect in every country. For short, semantically straightforward open-text responses, model size matters less for directional

conclusions than for precise agreement on individual classifications.

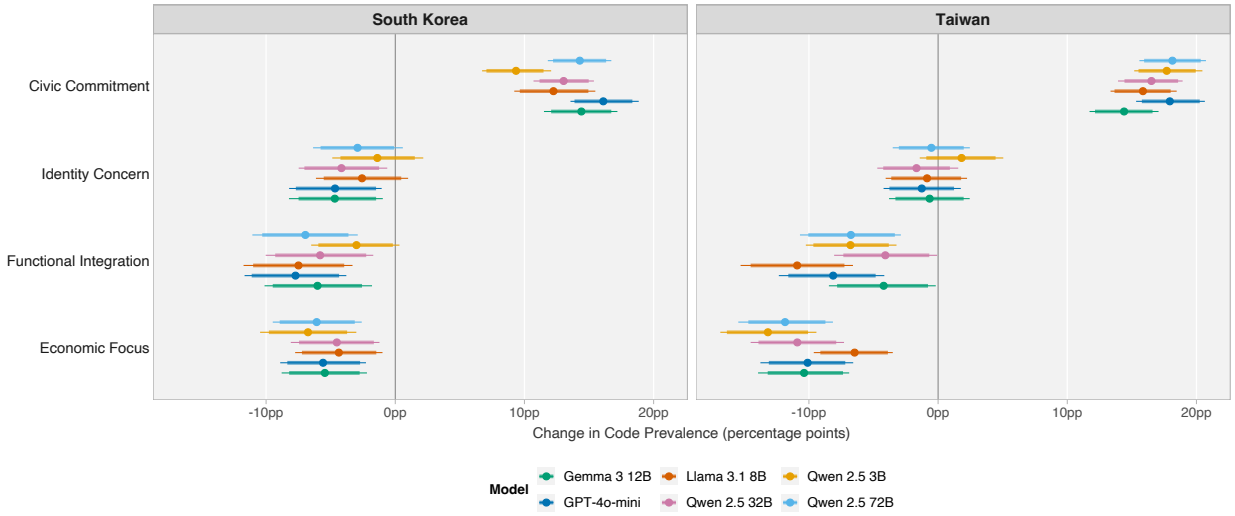

Figure 5: Cross-model robustness: cue effect estimates from six diverse LLMs. All models produce consistent treatment effect estimates for civic commitment.

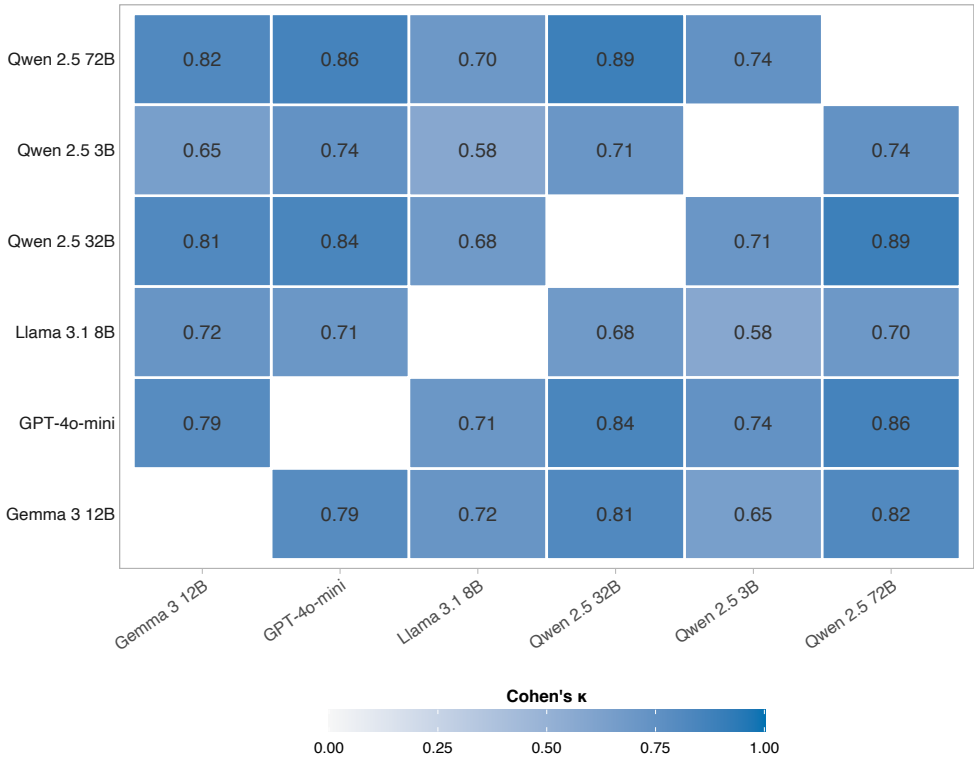

Figure 6: Cross-model agreement: pairwise Cohen's κ for six LLMs. Fleiss' multi-rater $\kappa = 0.745$.

Triangulation

The cleanest comparison between methods is for civic engagement, where a single STM topic maps directly to the LLM `civic_commitment` code in each country. In South Korea, the STM "Community Engagement" topic increases by 6.7 pp while the LLM civic code increases by 14.7 pp. In Taiwan, the STM "Community Participation & Identity" topic increases by 11.4 pp while the LLM civic code increases by 18.2 pp. All effects are significant. Both methods agree on the direction, significance, and cross-country pattern: the cue activates civic reasoning, and the effect is larger in Taiwan.

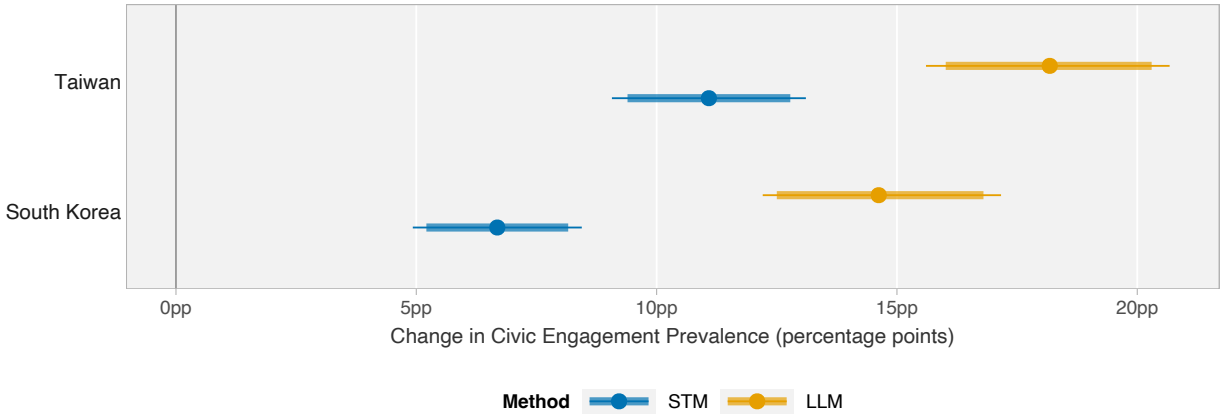

Figure 7: Triangulation: STM civic topic and LLM civic commitment treatment effects. The comparison where the topic-to-code mapping is unambiguous. Thick bars: 90% CIs; thin bars: 95% CIs.

Two factors explain why LLM estimates are larger than STM estimates. First, LLM codes are broader than individual STM topics: `civic_commitment` captures all civic reasoning in a single code, while STM distributes related content across multiple topics. Second, STM assigns topic proportions to every document regardless of content quality, while LLM classification routes uninformative responses to `other`, producing cleaner substantive codes and sharper treatment effects. This can be measured directly: the maximum topic probability ($\theta_{\max}$) for each document provides a measure of assignment clarity, and the median $\theta_{\max}$ is only 0.315 in South Korea and 0.303 in Taiwan (Supplementary Information, Appendix B). When diffusely assigned documents are progressively excluded, the STM civic effect rises monotonically and crosses the LLM estimate at $\theta_{\max} \approx 0.35$–$0.40$, which confirms that the gap between the two methods reflects how each handles uninformative text, not substantive disagreement about the

treatment effect.

For the remaining categories, the mapping between STM topics and LLM codes is not one-to-one, but the directional agreement is clear. On the practical and economic side, STM topics related to language, employment, and residence all decline when the cue is shown; the LLM similarly shows functional integration and economic focus declining. Identity concern is stable in both methods. Effects are consistently larger in Taiwan.

The v2 codebook refinement sharpens this convergence. By separating functional integration from civic commitment, the LLM classification now mirrors the STM finding that language and residence topics *decrease* when the cue is shown, which confirms that these practical attributes behave as substitutes for, not signals of, civic engagement.

The main analysis exploits within-design treatment randomization, a feature most conjoint experiments lack. Supplementary Information, Appendix D applies the framework to a standard conjoint on immigrant *admission* to South Korea [Denney and Green, 2021], where all attributes are always shown. Using LDA and two LLM classifiers (GPT-40-mini and Qwen 2.5 3B; $\kappa = 0.737$), the analysis shows that the framework transfers: LDA topics and LLM codes converge on the same constructs, and the approach retrospectively sharpens what the original study described broadly as "sociotropic" reasoning into three distinct constructs (sociotropic, economic, and functional integration).

Extension: Validation against Human-Coded Ground Truth

The main analysis relies on cross-model consensus because no human-coded ground truth exists for the citizenship conjoint responses. A natural question is how well LLM classifications agree with independent human coding. To test this, I apply the framework to a dataset where expert human classifications are available: Denney et al. [2024], a conjoint experiment on South Korean attitudes toward North Korean migrant entrepreneurship policy ($N = 2{,}009$). After the conjoint task, respondents answered an open-text question explaining their policy choice. Two human coders independently classified all responses into four categories: economic rationale, normative rationale, other rationale, and uninformative. Hu-

man inter-coder agreement was high (Cohen's $\kappa = 0.881$, 92.3% raw agreement), providing a reliable benchmark.

I classified the same responses using GPT-4o-mini and Llama 3.1 8B with a four-code codebook matching the human coding scheme. The LLMs classified English translations of the responses, matching the input the human coders used. I then compared LLM classifications to the human consensus. Because this validation uses English text, it confirms that the codebook and classification logic are well-calibrated but does not directly validate the multilingual classification pipeline that produces the main results.

The codebook required iteration. The initial version (v1) produced only fair agreement with the human coders (GPT-4o-mini $\kappa = 0.40$; Llama 3.1 8B $\kappa = 0.52$). Examining disagreements showed that both LLMs interpreted "economic" too narrowly and "uninformative" too broadly relative to the human coding. Because the conjoint is entirely about economic policy, the human coders classified policy-effectiveness language (e.g., "this policy would help them succeed") as economic, while the LLMs treated it as normative or uninformative. Broadening the economic category to include policy-effectiveness reasoning and tightening the uninformative code to cover only genuinely empty responses closed the gap across two rounds of revision.

Comparison	**Cohen's κ**	**Raw agreement**	**Landis & Koch**
Human–Human	0.881	92.3%	Almost perfect
GPT-4o-mini–Human	0.673	78.8%	Substantial
Llama 3.1 8B–Human	0.611	74.5%	Substantial
GPT-4o-mini–Llama 3.1 8B	0.717	81.5%	Substantial

Table 3: Inter-rater agreement for the NK entrepreneurship conjoint (v3 codebook, after two rounds of codebook revision). Human–human agreement from Denney et al. [2024].

After calibration, both LLMs fall in the "substantial" agreement range ($\kappa > 0.6$), and the two LLMs agree with each other at $\kappa = 0.717$. A gap remains between LLM-human and human-human agreement, but the trajectory across codebook versions is the more important finding: two rounds of refinement

moved GPT-4o-mini from $\kappa = 0.40$ (fair) to $\kappa = 0.67$ (substantial), a pattern that parallels the v1-to-v2 improvement in the main analysis. The codebook, not the model, is the primary determinant of classification quality.

This result has two implications for the main analysis. First, it provides external evidence that the iterative codebook workflow advocated in this paper produces meaningful gains in classification accuracy. Second, it calibrates expectations: even after careful iteration, LLM-human agreement ($\kappa \approx 0.6$–0.7) is lower than expert human-human agreement ($\kappa \approx 0.9$). The six-model cross-validation in the main analysis, which achieves Fleiss' $\kappa = 0.745$ across models of very different size and architecture, is consistent with this range and suggests that the multi-model consensus approach compensates for the gap between any single LLM and a human expert.

Discussion and Conclusion

Open-weight models produce results comparable to GPT-4o-mini, with all models agreeing on the direction and significance of treatment effects. Researchers need not depend on proprietary APIs, whose outputs may shift between model versions [Barrie et al., 2025], to conduct rigorous text classification. The open-weight models are the reproducible core; GPT-4o-mini confirms that these alternatives match the proprietary models used in current practice.

The model size results offer practical guidance. Agreement with GPT-4o-mini increases monotonically with model size within the Qwen 2.5 family, consistent with Chae and Davidson [2025]. Yet when each model's classifications are used individually, without any majority-vote aggregation, all six produce treatment effects in the same direction for every code in every country (Supplementary Information, Table C.6). The 3B model, runnable on a consumer GPU with 8GB VRAM, finds civic commitment increasing by +9.3 pp (South Korea) and +17.7 pp (Taiwan), compared to +16.1 and +17.9 pp for GPT-4o-mini. For researchers whose primary interest is directional treatment effects on short open-text responses, even small models suffice. Mid-range models (12B–32B) offer a reasonable tradeoff between cost and classification precision for more fine-grained applications.

The multi-model majority vote also offers a practical alternative to human-coded ground truth. Assembling expert coders for multilingual text is expensive and slow, and for many research questions the investment may not be justified when the goal is to validate a construct rather than produce a definitive coding of every response. When six models of different sizes, architectures, and training data agree on a classification, as they do for 93% of responses in this study, that consensus provides usable measurement even without human annotation. The ground-truth extension confirms that the gap between multi-model consensus and expert human coding is modest ($\kappa \approx 0.6$–0.7), and the codebook refinement process closes much of it. For researchers who lack access to expert coders in every language their data contains, cross-model validation is a scalable and defensible substitute.

This approach complements existing tools for assessing respondent engagement with conjoint profiles. Prior work has validated conjoint experiments against real-world behavior [Hainmueller et al., 2015] and examined visual attention using eye-tracking [Jenke et al., 2021]. Open-text responses provide a different window: how respondents *reason* about profile attributes, not just whether they attended to them. Unlike manipulation checks that condition on post-treatment responses and risk selection bias [Aronow et al., 2019], open-text validation examines the full sample. Other recent work scales open-ended responses on a continuous dimension [DiGiuseppe and Flynn, 2026]; the approach here is categorical, aimed at identifying what kind of reasoning respondents use and whether the treatment shifts it.

The iterative codebook design process makes a broader point: running an unsupervised model first can catch coding mistakes that would otherwise go unnoticed. The v1 codebook grouped civic engagement with language and residence under a single code. The STM showed that these move in opposite directions under the treatment: language and residence topics *decrease* when the cue is shown, so the v2 codebook separated them. That single revision (splitting functional integration from civic commitment) changed more classification outcomes than any other design choice in the analysis.

Substantively, the consistent cross-national difference in effect size deserves attention: the cue effects are larger in Taiwan than in South Korea across both methods and all classification models. This pattern may reflect differences in baseline expectations about immigrant civic participation, or Taiwan's more recent experience with large-scale immigration, which could make social integration cues more salient to

Taiwanese respondents than to South Korean respondents, who have had longer exposure to immigration policy debates.

Beyond these methodological and substantive contributions, several limitations should be noted. A fundamental caveat applies to any post-choice open-text elicitation: the question "Why did you choose this person?" invites post-hoc justification, not real-time introspection. Respondents may articulate reasons that are salient and socially acceptable rather than the considerations that actually drove their choice. The framework establishes that the treatment shifts what respondents *write*, which is necessary but not sufficient evidence that it shifts what they *think*. The substitution pattern, where civic cues displace language, residence, and economic reasoning rather than simply adding civic language, is harder to explain as mere recall, since it implies the cue changes the criteria respondents apply rather than just the words they use. Still, open-text validation is one source of evidence about construct processing, not a definitive window into cognition.

First, the open-text responses are short (median 6–10 characters). After morphological analysis and stopword removal, 41% of South Korean and 30% of Taiwanese documents contain just a single token. STM handles this through the Bayesian prior: documents with few words receive diffuse topic assignments dominated by the prior rather than the data, which is correct inference under uncertainty but produces conservative treatment effect estimates. Filtering short documents before estimation is inadvisable because document length may itself be affected by treatment (conditioning on a post-treatment variable) and because even single words can be informative (e.g., "volunteering" unambiguously signals civic engagement). This reflects a fundamental asymmetry between the two methods: STM requires a bag-of-words representation, which means preprocessing must strip grammatical structure, particles, and conjugation before estimation can proceed. LLM classification operates on the original text, retaining the linguistic context that preprocessing necessarily removes, and draws on world knowledge to classify even very short responses confidently. The $\theta_{\max}$ sensitivity analysis (Supplementary Information, Appendix B) quantifies how short-text dilution affects STM estimates. Researchers applying STM to similarly brief text should expect attenuated effects and consider reporting such sensitivity analyses. Second, open-text responses are self-generated and may not capture everything a respondent considered; some attributes

may have mattered but gone unwritten.

Third, the LLM codebook reflects particular theoretical commitments, and the v1-to-v2 iteration shows that initial codebook choices meaningfully shape results. There is a circularity in using STM results to both diagnose the codebook problem and validate the revised codebook on the same data; the LLM classification is fitted independently of the STM, which limits the concern, and the ground-truth extension validates the workflow on a separate dataset. Fourth, while six-model cross-validation provides reassurance, individual models vary in CJK text performance. Llama 3.1 8B inflates functional integration in Taiwan (43.7% vs. 30–37% for other models), probably due to weaker multilingual capabilities. Majority-vote aggregation mitigates but does not eliminate such model-specific biases.

Fifth, the framework has been tested on conjoint experiments about immigration in two East Asian democracies, using Korean and Traditional Chinese text. Several features of this application may not hold elsewhere. The responses are short and semantically straightforward, which favors LLM classification; longer, more discursive responses may introduce more ambiguity. The Qwen models used here were developed by Alibaba and perform well on CJK text; researchers working with Arabic, Swahili, or other languages with less representation in LLM training data should expect weaker multilingual performance and test accordingly. The conjoint design produces concrete, attribute-level reasoning that maps cleanly onto classification codes; constructs that are more abstract (democratic values, policy sophistication) may be harder to classify reliably. Whether the STM-to-LLM workflow transfers to other survey designs, other languages, and other substantive domains is an open question. Sixth, the approach assumes that open-text responses are genuinely human-generated. The data analyzed here pre-date widespread access to agentic AI tools: the primary surveys were fielded in 2024 and the ground-truth dataset comes from surveys conducted before ChatGPT's public release, so AI-generated contamination is not a concern for the present analysis. But for researchers adopting this pipeline going forward, the assumption deserves serious scrutiny.

Westwood [2025] demonstrate that agentic systems can complete surveys while passing 99.8% of standard attention checks, rendering conventional quality filters largely ineffective. That threat, outright deception by bot farms, is alarming but at least familiar; survey researchers have longstanding experience

managing data quality threats, even if the current tools require updating. The more unsettling challenge is delegation: real respondents who offload open-text answers to a language model rather than writing their own.[2] A respondent who delegates is not committing fraud; they are satisficing, outsourcing the cognitive effort of an open-ended question to a tool that produces fluent, plausible text. The result may be indistinguishable from a thoughtful human response. For an open-text validation framework, this is particularly corrosive: if respondents delegate their reasoning to an LLM, the analyst ends up using LLMs to classify LLM-generated text, and the exercise no longer validates anything about how people think.

There are no easy solutions. Detection-based approaches degrade as models improve, and treating delegation as a compliance problem places the burden on respondents rather than on instrument design. The more productive path is probably to design surveys where genuine participation requires less effort than delegation: shorter prompts, lower cognitive load, response formats that are easier to complete than to outsource. Pairing open-text questions with behavioral signals such as response timing or keystroke dynamics may also help, though these too face an arms race. The broader point is that open-text validation rests on a social contract, that the words on screen reflect a person's reasoning, and that contract is becoming harder to enforce. Researchers adopting this framework should treat the authenticity of open-text responses as an assumption to be actively verified, not taken for granted.

Combining structural topic modeling with LLM classification, and using each to inform the other, allows researchers to examine whether experimental manipulations shape respondent reasoning as intended. The primary application shows that the "cues of commitment" framework holds up in citizenship conjoint experiments across two East Asian democracies. Six diverse LLM classifiers and structural topic models converge: the social integration cue shifts respondent reasoning from practical and economic criteria toward active civic engagement. The iterative workflow, in which STM evidence exposed over-classification in the initial codebook and informed a more precise category structure, illustrates how the two methods can do more together than either does alone. The ground-truth validation against

[2] The distinction between deception and delegation as separate threats to survey research in the agentic era is developed by Yamil Velez, "Who Answered This Survey? Deception, Satisficing, and Delegation in the Agentic Era," *New Instruments* (Substack), March 30, 2026, `https://newinstruments.substack.com/p/who-answered-this-survey`.

human-coded data from a separate conjoint experiment confirms that iterative codebook refinement produces substantial LLM-human agreement ($\kappa \approx 0.6$–0.7) and that the codebook, not the model, is the primary lever for improving classification quality.

Researchers should consider embedding open-text questions in their survey experiments, even brief ones. The smallest open-weight model in this study (3B parameters) runs on a consumer GPU and produces directionally consistent results; larger models offer more precision at modest cost. The validity evidence is worth the modest additional burden of an open-text question.

Data Availability

Replication data, code, codebooks, and all analytical outputs are available in the replication archive at [URL/DOI to be added upon acceptance]. The archive includes the analysis-ready corpus, all LLM classification outputs, STM intermediate files, and publication figures.

Ethics Statement

This study is a secondary analysis of pre-existing survey data collected for other research projects [Denney et al., 2025, Denney and Green, 2021, Denney et al., 2024]. No new human subjects data were collected. The original surveys were conducted in accordance with applicable ethical standards and received appropriate institutional approvals.

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Competing Interests

The author declares none.

Use of AI-Assisted Tools

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References

Peter M. Aronow, Jonathon Baron, and Lauren Pinson. A note on dropping experimental subjects who fail a manipulation check. *Political Analysis*, 27(4):572–589, 2019. https://doi.org/10.1017/pan.2019.5.

Christopher A. Bail. Can generative AI improve social science? *Proceedings of the National Academy of Sciences*, 121(21):e2314021121, 2024. https://doi.org/10.1073/pnas.2314021121.

Kirk Bansak, Jens Hainmueller, Daniel J. Hopkins, and Teppei Yamamoto. Conjoint survey experiments. In James N. Druckman and Donald P. Green, editors, *Advances in Experimental Political Science*, pages 19–41. Cambridge University Press, 2021.

Christopher Barrie, Alexis Palmer, and Arthur Spirling. Replication for language models: Problems, principles, and best practices for political science. Working paper, 2025.

Kenneth Benoit, Kohei Watanabe, Haiyan Wang, Paul Nulty, Adam Obeng, Stefan Müller, and Akitaka Matsuo. quanteda: An R package for the quantitative analysis of textual data. *Journal of Open Source Software*, 3(30):774, 2018. https://doi.org/10.21105/joss.00774.

Kenneth Benoit, Scott De Marchi, Conor Laver, Michael Laver, and Jinshuai Ma. Using large language models to analyze political texts through natural language understanding. *American Journal of Political Science*, 2026. https://doi.org/10.1111/ajps.70050.

David M. Blei, Andrew Y. Ng, and Michael I. Jordan. Latent Dirichlet allocation. *Journal of Machine Learning Research*, 3:993–1022, 2003. Available at https://jmlr.org/papers/v3/blei03a.html.

Youngjin Chae and Thomas Davidson. Large language models for text classification: From zero-shot learning to instruction-tuning. *Sociological Methods & Research*, 2025. https://doi.org/10.1177/00491241251325243.

Steven Denney and Christopher Green. Who should be admitted? conjoint analysis of South Korean attitudes toward immigrants. *Ethnicities*, 21(1):120–145, 2021. https://doi.org/10.1177/1468796820916609.

Steven Denney, Peter Ward, and Christopher Green. Public support for migrant entrepreneurship: The case of North Koreans in the Republic of Korea. *International Migration Review*, 58(2):781–805, 2024. https://doi.org/10.1177/01979183231203927.

Steven Denney, Nicholas A.R. Fraser, and H. Christoph Steinhardt. Cues of commitment: Civic engagement and attitudes toward immigrant naturalization in East Asia. Working paper. Available at https://github.com/scdenney/cues-east-asia, 2025.

Matthew J. Denny and Arthur Spirling. Text preprocessing for unsupervised learning: Why it matters, when it misleads, and what to do about it. *Political Analysis*, 26(2):168–189, 2018. https://doi.org/10.1017/pan.2017.44.

Matthew R. DiGiuseppe and Michael E. Flynn. Scaling open-ended survey responses using LLM-paired comparisons. *Public Opinion Quarterly*, 2026. https://doi.org/10.1093/poq/nfag013.

Naoki Egami, Christian J. Fong, Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. How to make causal inferences using texts. *Science Advances*, 8(42), 2022. https://doi.org/10.1126/sciadv.abg2652.

Christian Fong and Justin Grimmer. Causal inference with latent treatments. *American Journal of Political Science*, 67(2):374–389, 2023. https://doi.org/10.1111/ajps.12649.

John G. Geer. What do open-ended questions measure? *Public Opinion Quarterly*, 52(3):365–367, 1988. https://doi.org/10.1086/269113.

Matthew Gentzkow, Bryan Kelly, and Matt Taddy. Text as data. *Journal of Economic Literature*, 57(3):535–574, 2019. https://doi.org/10.1257/jel.20181020.

Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30):e2305016120, 2023. https://doi.org/10.1073/pnas.2305016120.

Rebecca A. Glazier, Amber E. Boydstun, and Jessica T. Feezell. Self-coding: A method to assess semantic validity and bias when coding open-ended responses. *Research & Politics*, 8(3):1–8, 2021. https://doi.org/10.1177/20531680211031752.

Justin Grimmer and Brandon M. Stewart. Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297, 2013. https://doi.org/10.1093/pan/mps028.

Justin Grimmer, Margaret E. Roberts, and Brandon M. Stewart. *Text as Data: A New Framework for Machine Learning and the Social Sciences*. Princeton University Press, 2022.

Ingar Haaland, Christopher Roth, Stefanie Stantcheva, and Johannes Wohlfart. Understanding economic behavior using open-ended survey data. *Journal of Economic Literature*, 63(4):1244–1280, 2025. https://doi.org/10.1257/jel.20251780.

Jens Hainmueller and Daniel J. Hopkins. The hidden American immigration consensus: A conjoint analysis of attitudes toward immigrants. *American Journal of Political Science*, 59(3):529–548, 2015. https://doi.org/10.1111/ajps.12138.

Jens Hainmueller, Daniel J. Hopkins, and Teppei Yamamoto. Causal inference in conjoint analysis: Understanding multidimensional choices via stated preference experiments. *Political Analysis*, 22(1):1–30, 2014. https://doi.org/10.1093/pan/mpt024.

Jens Hainmueller, Dominik Hangartner, and Teppei Yamamoto. Validating vignette and conjoint survey experiments against real-world behavior. *Proceedings of the National Academy of Sciences*, 112(8):2395–2400, 2015. https://doi.org/10.1073/pnas.1416587112.

Andrew Halterman and Katherine A. Keith. Codebook LLMs: Evaluating LLMs as measurement tools for political science concepts. *Political Analysis*, 2025. https://doi.org/10.1017/pan.2025.10017.

Michael Heseltine and Bernhard Clemm von Hohenberg. Large language models as a substitute for human experts in annotating political text. *Research & Politics*, 11(1):1–10, 2024. https://doi.org/10.1177/20531680241236239.

William Hobbs and Jon Green. Categorizing topics versus inferring attitudes: A theory and method for analyzing open-ended survey responses. *Political Analysis*, 33:231–251, 2025. https://doi.org/10.1017/pan.2024.23.

Libby Jenke, Kirk Bansak, Jens Hainmueller, and Dominik Hangartner. Using eye-tracking to understand decision-making in conjoint experiments. *Political Analysis*, 29(1):75–101, 2021. https://doi.org/10.1017/pan.2020.11.

Gaël Le Mens and Aina Gallego. Positioning political texts with large language models by asking and averaging. *Political Analysis*, 33(3):274–282, 2025. https://doi.org/10.1017/pan.2024.29.

Christopher Lucas, Richard A. Nielsen, Margaret E. Roberts, Brandon M. Stewart, Alex Storer, and Dustin Tingley. Computer-assisted text analysis for comparative politics. *Political Analysis*, 23(2):254–277, 2015. https://doi.org/10.1093/pan/mpu019.

Arthur Lupia. How to improve coding for open-ended survey data: Lessons from the ANES. In David L. Vannette and Jon A. Krosnick, editors, *The Palgrave Handbook of Survey Research*, pages 121–126. Springer, 2018. https://doi.org/10.1007/978-3-319-54395-6_16.

Jonathan Mellon, Jack Bailey, Ralph Scott, James Breckwoldt, Marta Miori, and Phillip Schmedeman. Do AIs know what the most important issue is? Using language models to code open-text social survey responses at scale. *Research & Politics*, 11(1), 2024. https://doi.org/10.1177/20531680241231468.

Joseph T. Ornstein, Elise N. Blasingame, and Jake S. Truscott. How to train your stochastic parrot: Large language models for political texts. *Political Science Research and Methods*, 13:264–281, 2025. https://doi.org/10.1017/psrm.2024.64.

Steve Rathje, Dan-Mircea Mirea, Ilia Sucholutsky, Raja Marjieh, Claire Robertson, and Jay J. Van Bavel. GPT is an effective tool for multilingual psychological text analysis. *Proceedings of the National Academy of Sciences*, 121(34):e2308950121, 2024. https://doi.org/10.1073/pnas.2308950121.

Margaret E. Roberts, Brandon M. Stewart, Dustin Tingley, Christopher Lucas, Jetson Leder-Luis, Shana Kushner Gadarian, Bethany Albertson, and David G. Rand. Structural topic models for open-ended survey responses. *American Journal of Political Science*, 58(4):1064–1082, 2014. https://doi.org/10.1111/ajps.12103.

Margaret E. Roberts, Brandon M. Stewart, and Edoardo M. Airoldi. A model of text for experimentation in the social sciences. *Journal of the American Statistical Association*, 111(515):988–1003, 2016. https://doi.org/10.1080/01621459.2016.1141684.

Margaret E. Roberts, Brandon M. Stewart, and Dustin Tingley. stm: An R package for structural topic models. *Journal of Statistical Software*, 91(2):1–40, 2019. https://doi.org/10.18637/jss.v091.i02.

Petter Törnberg. Large language models outperform expert coders and supervised classifiers at annotating political social media messages. *Social Science Computer Review*, 43(6):1181–1195, 2025. https://doi.org/10.1177/08944393241286471.

Sean J. Westwood. The potential existential threat of large language models to online survey research. *Proceedings of the National Academy of Sciences*, 2025. https://doi.org/10.1073/pnas.2518075122.

Jeffrey Ziegler. A text-as-data approach for using open-ended responses as manipulation checks. *Political Analysis*, 30(2):289–297, 2022. https://doi.org/10.1017/pan.2021.2.

Caleb Ziems, William Held, Omar Shaber, Jiaao Lu, Mengzhou Li, and Diyi Yang. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291, 2024. https://doi.org/10.1162/coli_a_00502.